

ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:	A. Mounish	Reg. No.:	192472273
Section / Batch:	CSE AI	Date:	20/08/26

Instructions to Students

- Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.
 - Compare all four models: BERT, GPT, T5, and LLaMA.
 - Explain the **architecture and working principle** of each model.
 - Compare their **pre-training objectives and capabilities**.
 - Map each model to appropriate **real-world applications**.
 - Identify the **strengths and limitations** of each model.
-
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.

- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I A. Mounish / 192472273 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student:

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

ARCHITECTURE-BASED COMPARISON

7. Compare the attention mechanisms and contextual understanding capabilities of BERT, GPT, T5, and LLaMA. Discuss their suitability for different NLP applications.

1. Question Requirement

Compare the attention mechanisms and contextual understanding capabilities of BERT, GPT, T5, and LLaMA. Discuss their suitability for different NLP applications.

2. Introduction

BERT, GPT, T5, and LLaMA are based on the Transformer family of neural network architectures. However, they are not identical. BERT uses an encoder-only structure and is mainly designed for language understanding. GPT and LLaMA use decoder-only structures and are strongly oriented toward autoregressive generation. T5 uses an encoder-decoder structure and frames many NLP problems as text-to-text transformations.

The attention mechanism is the key concept behind this comparison. Attention allows the model to decide which tokens are important when constructing a representation. The direction and type of attention determine how much context the model can use and how it produces its output.

3. What Is Attention?

Attention is a mechanism that allows a neural language model to assign different importance to tokens in a sequence. Instead of treating every word equally, the model calculates relationships between tokens and uses the most relevant information when constructing contextual representations.

Self-attention allows tokens in the same sequence to interact with one another. In Transformer models, this mechanism helps capture relationships such as subject–verb agreement, references between words, and connections between important terms appearing far apart in a sentence.

4. BERT – Detailed Explanation

4.1 Architecture

BERT stands for Bidirectional Encoder Representations from Transformers. It uses an encoder-only Transformer architecture. The input sentence is converted into tokens, and these tokens pass through a stack of Transformer encoder layers. Each layer contains multi-head self-attention and feed-forward processing.

4.2 Attention Mechanism

BERT uses bidirectional self-attention. This means that when the model builds a representation for a word, it can use information from words occurring before and after that word. The bidirectional design is particularly useful for understanding a word according to its complete sentence context.

4.3 Contextual Understanding

BERT creates contextual representations rather than giving every occurrence of a word the same representation. Consider the word “bank” in two sentences: “I deposited money in the bank” and “We sat near the river bank.” The surrounding words provide different context, allowing BERT to distinguish the intended meaning.

4.4 Step-by-Step Working

- Tokenize the input sentence into tokens.
- Convert tokens into input representations.
- Pass the sequence through Transformer encoder layers.
- Apply bidirectional self-attention to relate each token to surrounding tokens.
- Create contextual representations for the tokens.
- Use the representations for a downstream NLP task such as classification or NER.

4.5 Example

For the sentence “The student submitted the assignment because it was due today,” BERT can use the surrounding context to understand that “it” refers to the assignment. This type of contextual relationship is important for language-understanding tasks.

4.6 Suitable Applications

- Sentiment analysis
- Text classification
- Named Entity Recognition (NER)
- Extractive question answering
- Semantic similarity and language understanding

4.7 Advantages

- Strong bidirectional contextual representations.
- Very effective for understanding-oriented NLP tasks.
- Useful for token-level tasks such as NER.
- Can be fine-tuned for many downstream applications.

4.8 Limitations

- Not primarily designed for open-ended text generation.
- Usually adapted to specific tasks through fine-tuning.
- Generating long free-form responses is not its main purpose.

BERT - Bidirectional Context

Encoder-only Transformer | understanding-oriented

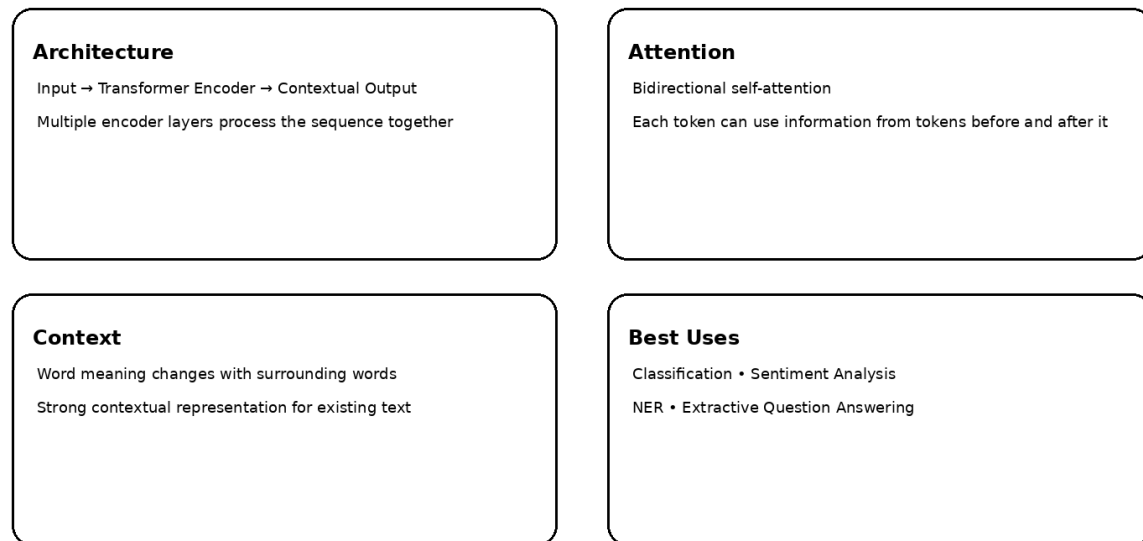


Figure 1. BERT visual summary: encoder-only architecture, bidirectional attention, contextual understanding, and applications.

5. GPT – Detailed Explanation

5.1 Architecture

GPT uses a decoder-only Transformer architecture. It is an autoregressive language model, meaning that it generates a sequence one token at a time. The output from one step becomes part of the context used for the next step.

5.2 Attention Mechanism

GPT uses causal self-attention, also called masked self-attention. During next-token prediction, a position cannot directly attend to future tokens. This maintains the left-to-right generation process and prevents the model from using information that would not yet be available during generation.

5.3 Contextual Understanding

GPT uses the preceding prompt and generated tokens as context. As the sequence becomes longer, the model can use earlier information to determine what should be generated next. This makes it suitable for dialogue, instruction following, content generation, and other generative applications.

5.4 Step-by-Step Working

- Tokenize the prompt.
- Represent the input tokens inside the decoder stack.
- Apply causal self-attention so future positions are hidden.

- Calculate probabilities for the next token.
- Select or sample the next token.
- Add the generated token to the context and repeat until the response is complete.

5.5 Example

If the prompt is “Artificial intelligence is useful because...”, GPT uses the preceding words to predict a suitable continuation. In a chatbot, the conversation history becomes part of the available context for generating the next response.

5.6 Suitable Applications

- AI chatbots and virtual assistants
- Text and content generation
- Generative question answering
- Story and email generation
- Code generation
- Conversational systems

5.7 Advantages

- Strong open-ended text generation.
- Flexible prompting across many tasks.
- Suitable for conversational applications.
- Can generate explanations, summaries, and code.

5.8 Limitations

- Generated content can contain incorrect information.
- Large models can require substantial computing resources.
- Its causal context is different from BERT's bidirectional understanding.

GPT - Autoregressive Generation

Decoder-only Transformer | generation-oriented

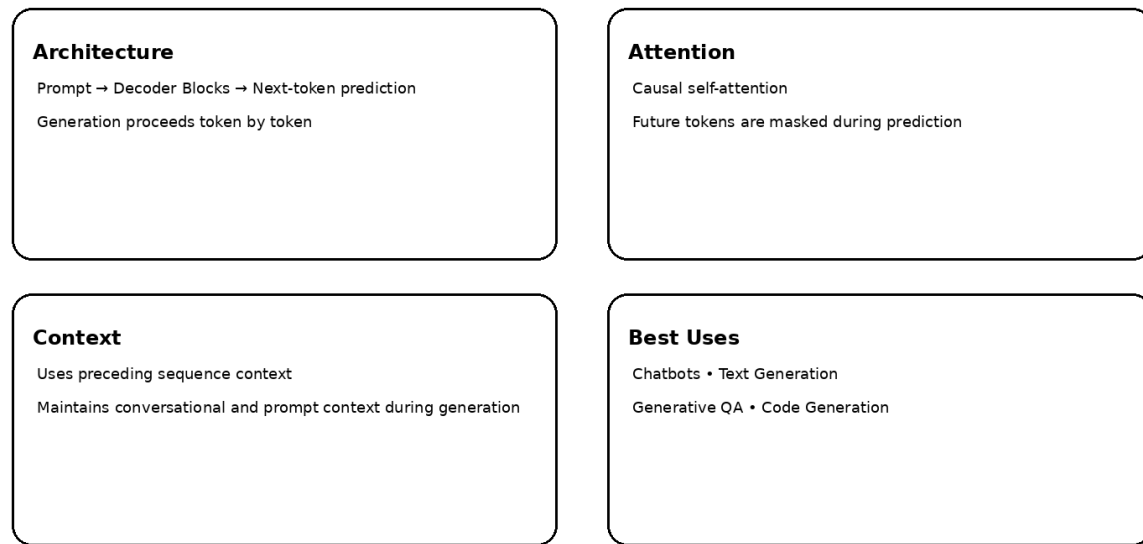


Figure 2. GPT visual summary: decoder-only architecture, causal attention, autoregressive generation, and applications.

6. T5 – Detailed Explanation

6.1 Architecture

T5 stands for Text-To-Text Transfer Transformer. It uses an encoder-decoder Transformer architecture. The encoder reads the input sequence and creates a contextual representation. The decoder then generates the output sequence.

6.2 Attention Mechanism

T5 combines encoder self-attention, decoder self-attention, and encoder-decoder cross-attention. The encoder self-attention allows the input to be processed using its context. During generation, the decoder uses its causal self-attention and also uses cross-attention to focus on relevant encoder information.

6.3 Contextual Understanding

The encoder can process the input as a complete sequence, allowing it to build a rich representation of the source text. The decoder then uses that representation while generating the output. This makes T5 especially suitable for tasks where one text needs to be converted into another text.

6.4 Step-by-Step Working

- Convert the task and source text into a text-to-text input.
- Pass the input through the encoder.

- Use encoder self-attention to build contextual representations.
- Pass the encoder representation to the decoder through cross-attention.
- Generate the required output token by token.
- Continue until the output sequence is complete.

6.5 Example

For summarization, the input can be formatted as “summarize: [long article].” The encoder represents the article and the decoder generates a shorter summary. For translation, an input such as “translate English to French: Good morning” can produce “Bonjour.”

6.6 Suitable Applications

- Machine translation
- Text summarization
- Question answering
- Text classification in a text-to-text format
- Paraphrasing and text transformation

6.7 Advantages

- Unified text-to-text framework.
- Combines input understanding with output generation.
- Suitable for many sequence-to-sequence tasks.
- Flexible task formulation.

6.8 Limitations

- Encoder-decoder processing can be more complex than a single-stack architecture.
- Requires appropriate task formulation and training/fine-tuning.
- Can require substantial computation for larger configurations.

T5 - Text-to-Text Transformation

Encoder-decoder Transformer | transformation-oriented

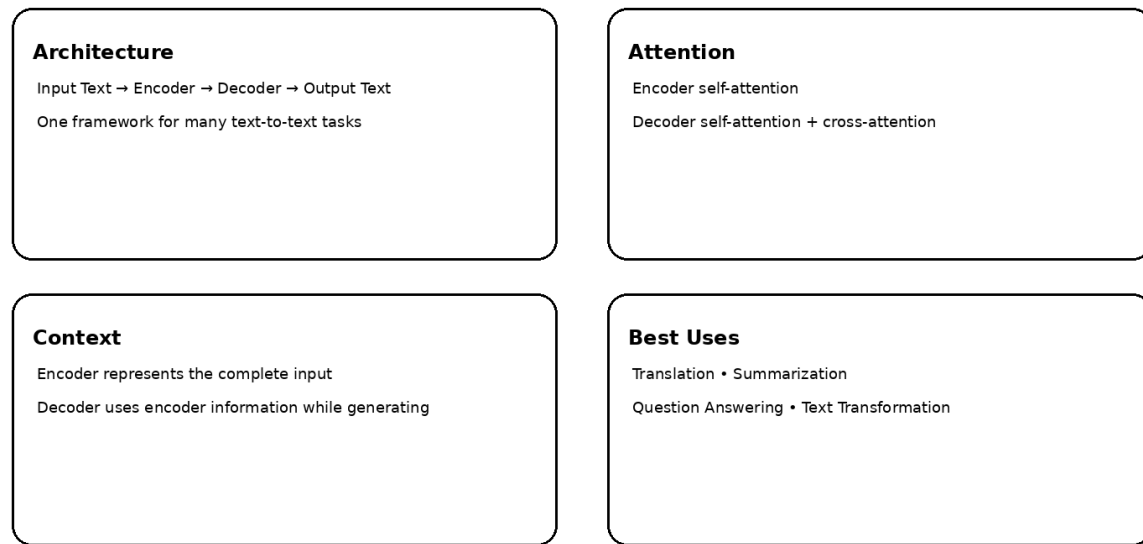


Figure 3. T5 visual summary: encoder-decoder architecture, self-attention, cross-attention, and text-to-text applications.

7. LLaMA – Detailed Explanation

7.1 Architecture

LLaMA is a family of large language models based on a decoder-only Transformer architecture. Like other autoregressive language models, it processes the available context and generates output sequentially.

7.2 Attention Mechanism

LLaMA uses causal self-attention. Each position uses the available preceding context rather than future tokens. This supports autoregressive next-token prediction and makes the architecture naturally suited to text generation.

7.3 Contextual Understanding

LLaMA can use the prompt and preceding generated tokens as context. This supports multi-turn conversations, explanations, long-form generation, question answering, summarization, and other generative applications, subject to the model's context capacity and training.

7.4 Step-by-Step Working

- Tokenize the input prompt.
- Pass tokens through decoder Transformer layers.
- Apply causal self-attention to the available context.

- Predict the next token.
- Append the selected token to the sequence.
- Repeat the process to generate the response.

7.5 Example

If a user asks, “Explain cloud computing in simple words,” LLaMA uses the prompt context to generate a natural-language explanation. In a chatbot, previous conversation turns can be included in the prompt so that the response remains relevant to the discussion.

7.6 Suitable Applications

- Generative AI assistants
- Chatbots
- Question answering
- Text generation
- Summarization
- Code generation and code assistance

7.7 Advantages

- Strong general-purpose generation capability.
- Useful for conversational and instruction-oriented applications.
- Can be adapted for specialized generative tasks.

7.8 Limitations

- Large model variants can require substantial memory and computation.
- Generated information can be incorrect and requires verification.
- Causal generation differs from bidirectional understanding in encoder-only models.

LLaMA - Generative Language Model

Decoder-only Transformer | general-purpose generation

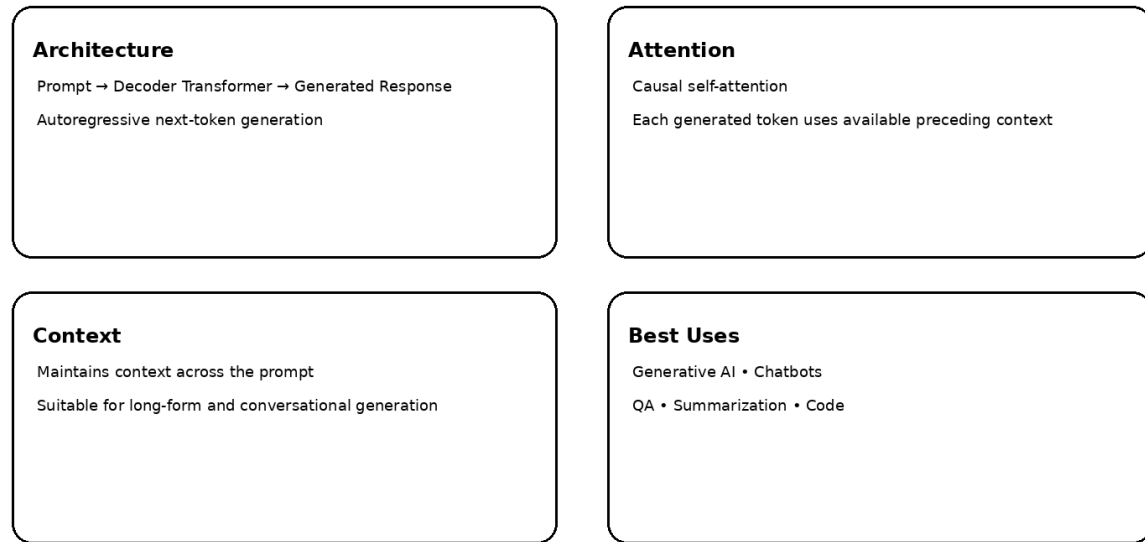


Figure 4. LLaMA visual summary: decoder-only architecture, causal attention, contextual generation, and applications.

8. Direct Comparison of the Four Models

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Attention	Bidirectional self-attention	Causal self-attention	Self + cross-attention	Causal self-attention
Context	Both sides of input	Preceding context	Full encoded input + decoder context	Preceding context
Main purpose	Understanding	Generation	Text-to-text	General generation
Strong areas	Classification, NER, extractive QA	Chatbots, generation, code	Translation, summarization	Chatbots, generation, assistants

9. NLP Application Suitability

The choice of model depends on the task. Understanding-oriented tasks generally benefit from encoder-style contextual representations, while open-ended generation benefits from autoregressive decoder architectures. Encoder-decoder models are particularly useful when the application transforms one sequence into another.

Which Model Fits Which NLP Task?

Qualitative suitability based on architecture and intended use

Language understanding	BERT
Classification	BERT
NER	BERT
Chatbots	GPT / LLaMA
Text generation	GPT / LLaMA
Translation	T5
Summarization	T5 / GPT / LLaMA
Extractive QA	BERT

Figure 5. Qualitative application map showing which model families are naturally suited to representative NLP tasks.

10. Strengths and Limitations – Overall Comparison

Model	Strengths	Limitations
BERT	Bidirectional contextual understanding; strong classification and NER.	Limited open-ended generation.
GPT	Strong generation; conversational capability; flexible prompting.	Possible factual errors; computational cost for large models.
T5	Strong input-to-output transformation; unified text-to-text framework.	More involved encoder-decoder processing.
LLaMA	Strong general-purpose generation; broad generative applications.	Resource requirements increase with model size; output verification needed.

11. Final Conclusion

BERT, GPT, T5, and LLaMA all use the Transformer concept, but their attention mechanisms determine how they use context and what type of NLP task they handle best. BERT uses bidirectional attention and is strongest for language understanding. GPT and LLaMA use causal attention and are strong for autoregressive generation. T5 combines an encoder and decoder with cross-attention and is particularly effective for text-to-text transformation.

For classification, NER, and extractive question answering, BERT is a strong choice. For chatbots and open-ended generation, GPT or LLaMA is more suitable. For translation, summarization, and sequence transformation, T5 is especially appropriate. Therefore, the correct model depends on the requirements of the NLP application rather than on choosing one model as universally best.